

Vdbench: A Storage Benchmarking Tool

Considered one of the most versatile of benchmarks, Vdbench can be used as a GUI tool or a CLI tool. It is highly portable and works on different platforms like Windows, Solaris, Linux, AIX, OS/X and HP/UX.

Vdbench is a controlled I/O load simulation tool for storage systems. It gives control over workload parameters like I/O rate, LUN or file size, transfer size and cache hit percentage. It generates reports which are Web accessible in HTML format. It is an open source tool from Oracle, with 99 per cent of it written in Java and 1 per cent in C language. It works on all Linux and Windows operating systems. It is designed to execute workloads on a storage system.

In a simple equation, $f(\text{workload}, \text{Config}) = \text{Power} + \text{Performance}$. With a set of workloads defined through a configuration file, you can measure the performance of any storage array. It runs as two or more Java virtual machines. The one we invoke is the master, which takes care of parsing and reporting. The actual workload is executed by one or more slaves. A slave can run along with the master or in a different host. Vdbench can be useful for identifying data corruption on the storage system. A combination of data validation and journaling allows you to identify data corruption issues across executions. In short, it is a truly enterprise-grade storage benchmark that is free and open source.

Installation

Vdbench can be downloaded from <http://www.oracle.com/technetwork/server-storage/vdbench-downloads-1901681.html>

The `vdbench50405.zip` file contains the executables needed for Windows and Linux environments. Just extract the zip file and run `./vdbench -t (Raw workload)` or `./vdbench -tf (File system workload)` to verify the installation. Results will be stored under `output/summary.html`.

Parameter definition

Vdbench has three basic definitions in the configuration file.

1. SD or storage definition: This defines what storage to use for the testing.
2. WD or workload definition: This defines the workload parameter for the testing.
3. RD or run definition: This defines storage, workload and run duration.

Storage definition

SD defines the storage used for the testing. Make sure you select the right storage or else everything on the selected disk could get destroyed. The SD name should be unique.

```
sd=default, size=100g
sd=sd1, lun=/dev/sdb0
sd=sd2, lun=/dev/sdb1, size=100g
sd=sd3, lun=/dev/sdb2, size=200g
```

Workload definition

WD defines the workload used by the script for testing. The

